

Heat Insert Tool Safety Policy

1. Purpose

This policy defines the mandatory safety rules for using heat insert tools, heat-set insert presses, soldering-iron-style insertion tips, and related fixtures in the lab. It is intended to prevent burns, fire, electrical shock, fume exposure, chemical exposure, eye injury, pinch injury, cuts, equipment damage, and property damage.

Every employee, contractor, intern, visitor, or other person who operates, assists with, cleans, maintains, or observes use of the heat insert tool must follow this policy and sign the acknowledgement before participating in heat insert work.

2. Scope

This policy applies to:

- Installing heat-set threaded inserts into approved plastic parts using a heated insertion tool, insertion press, or soldering-iron-style insert tip.
- Setup, temperature selection, fixture use, insertion, cooling, cleanup, troubleshooting, and routine maintenance of heat insert equipment.
- Handling hot tips, hot press heads, hot stands, hot fixtures, metal inserts, softened plastic, failed parts, and related hand tools.
- Anyone working near the heat insert station while equipment is powered, heating, in use, cooling, being cleaned, or being serviced.

This policy does not cover oven reflow, oven curing, bake-out, or thermal chamber work, which are covered by the Oven Safety Policy. It does not replace the Soldering Iron Safety Policy for hand soldering or the Bambu Lab P2S 3D Printer Safety Policy for 3D printing. When a heat insert operation overlaps with those processes, the more specific policy and the more conservative control must be followed.

3. Conservative Safety Assumption

The person handling the heat insert tool shall treat every tip, press head, stand, fixture, insert, nearby part surface, and recently processed part as hot unless they have personally verified that it is cool.

The operator shall treat heat insert work as a combined heat, fire, electrical, fume, pinch, sharp-object, and ergonomic hazard. Familiarity with routine insert installation does not remove the need for

PPE, ventilation, housekeeping, and active attention.

If the tool is damaged, unstable, producing smoke from the handle or cord, overheating abnormally, failing to regulate temperature, missing its stand or holder, or producing an abnormal odor, it must be removed from service until inspected and cleared.

4. Main Ways a Person Could Be Hurt

Employees must understand that injury can occur during setup, insertion, part handling, cooldown, cleanup, and troubleshooting. The principal hazards are:

1. **Burns from hot tools and inserts.** Insert tips, press heads, stands, fixtures, metal inserts, and recently processed parts can burn skin.
2. **Burns from softened plastic.** Softened or melted plastic can stick to skin and continue burning.
3. **Eye injury.** Inserts can slip or eject, parts can crack, tools can slip, and plastic fragments can fly during insertion or failed part removal.
4. **Fume exposure.** Heated plastic, residue, adhesive, printed-material additives, and overheated parts can release irritating or hazardous fumes, odors, smoke, and particulates.
5. **Fire.** Hot tips can ignite paper, cardboard, wipes, packaging, plastic scraps, solvents, aerosols, clothing, dust, or other combustible material.
6. **Electrical shock.** Damaged cords, wet surfaces, exposed wiring, improper repairs, modified controllers, or liquids near powered tools can create shock hazards.
7. **Pinch, crush, or puncture injury.** Insertion presses, clamps, fixtures, tweezers, pliers, inserts, sharp part edges, and broken plastic can pinch, cut, puncture, or scrape skin.
8. **Unexpected process behavior.** Wrong temperature, wrong dwell time, wrong insert size, wrong hole size, wrong material, or excessive force can cause part failure, smoke, tool slip, or burns.
9. **Ergonomic strain.** Pressing inserts, holding small parts, repeated fine work, and awkward wrist position can cause strain or fatigue.
10. **Complacency risk.** Injuries are more likely when an operator assumes a part has cooled, leaves a hot tool loose on the bench, skips ventilation, uses unknown materials, or rushes insertion work.

5. Operator Responsibilities

5.1 Person Handling the Heat Insert Tool

The person handling the heat insert tool shall:

- Complete required training before use.

- Follow this policy, the manufacturer's instructions, the lab operating procedure, and applicable material safety data sheets.
- Review and follow the Soldering Iron Safety Policy if the heat insert setup uses a soldering-iron-style heater, tip, stand, or fume-control setup.
- Review and follow the Bambu Lab P2S 3D Printer Safety Policy when installing inserts into 3D-printed parts.
- Confirm the work area, tool, power cord, controller, stand, insert tip, temperature setting, material, fixture, ventilation, and PPE are safe before use.
- Use fume extraction or local ventilation when heating plastic or any material that produces odor or fumes.
- Keep combustible materials and liquids away from hot tools.
- Keep the hot tool in its stand, holder, or press whenever it is not actively in use.
- Stop work immediately if unsafe conditions exist.
- Report injuries, near misses, smoke events, fire events, electrical faults, abnormal odors, damaged equipment, temperature-control failures, ventilation failures, or unsafe conditions.
- Remove unsafe equipment from service.

5.2 Observers and Visitors

Observers and visitors may approach the heat insert station only with permission from the person handling the equipment. They must avoid hot tools, hot parts, fumes, sharp edges, moving press parts, and active work areas. Visitors may not operate the heat insert tool unless trained and authorized.

6. Authorization and Training

No person may operate the heat insert tool independently until all of the following are complete:

- Review of this policy and signature of the acknowledgement.
- Review of the current manufacturer manual and lab operating procedure.
- Review of the Soldering Iron Safety Policy if the tool uses soldering-iron-style components.
- Review of the Bambu Lab P2S 3D Printer Safety Policy before installing inserts into 3D-printed parts.
- Hands-on training by a person already competent with the specific tool, materials, fixtures, and procedure.
- Demonstrated ability to inspect the tool, verify ventilation, set a safe temperature, select the correct insert and tip, secure the part, install inserts without unsafe force, handle hot parts, shut down the tool, clean the area, and respond to burns, fumes, smoke, or fire.
- Demonstrated understanding of hot surfaces, material limits, fume hazards, fire prevention, electrical hazards, required PPE, and incident reporting.

Refresher review is required after any incident, near miss, equipment change, material change, process change, or extended period without use.

7. Approved Materials and Ventilation

Only approved inserts, tips, fixtures, and plastic materials may be used. The operator must confirm the material identity, softening temperature, recommended insertion temperature, required ventilation, and compatibility with the tool before heating.

Fume extraction or local ventilation must be used whenever heating plastic, adhesive, contaminated parts, unknown residues, or any material that produces odor, smoke, fumes, or particulates.

Ventilation must be positioned so emissions are captured before entering the breathing zone.

The following are prohibited unless specifically approved by the person responsible for the lab or equipment:

- Installing inserts into unknown, unlabeled, contaminated, wet, oily, chemically treated, visibly degraded, or incompatible materials.
- Heating materials above their rated temperature, decomposition temperature, manufacturer limit, or approved lab process limit.
- Heating PVC, vinyl, halogenated plastics, unknown plastics, unknown coatings, unknown adhesives, batteries, sealed containers, or aerosol-contaminated items.
- Using the heat insert tool on parts that require ventilation, filtration, fixturing, or fire controls the lab setup cannot safely provide.
- Using unapproved insert tips, loose tips, damaged fixtures, or improvised holders.

If a material produces strong odor, visible smoke, charring, bubbling, flame, unexpected melting, or abnormal reaction, stop the process if safe to do so and report the issue.

8. Required Engineering Controls

The following controls must be in place before heat insert work:

- A stable, temperature-controlled heat insert tool or heat-set insert press suitable for the task.
- A proper stand, holder, guard, or press fixture for the hot tool.
- A stable, level, heat-resistant, nonflammable work surface.
- Fixtures, clamps, or holders that keep hands out of the hot tip and press path where needed.
- Fume extraction or local ventilation when required by this policy, the material, or the lab procedure.
- Adequate lighting and a clear bench.
- Safe cable routing so cords cannot pull a hot tool off the bench or create a trip hazard.
- A suitable fire extinguisher accessible in the lab.

- Approved containers for failed parts, inserts, contaminated wipes, sharp debris, and chemical waste.

If any required control is missing or defective, heat insert work must not begin.

9. Required PPE

Minimum PPE for normal heat insert work:

- Safety glasses.
- Heat-resistant finger protection or gloves when handling hot inserts, hot fixtures, or recently processed parts.
- Nitrile gloves when handling contaminated parts, adhesives, solvents, uncured chemicals, or cleaning chemicals.
- Cut-resistant gloves when handling sharp fixtures, broken parts, or sharp plastic if they do not create an entanglement or dexterity hazard.

Operators must secure long hair, loose clothing, jewelry, lanyards, hoodie strings, and dangling items before using the heat insert tool.

PPE does not make it safe to touch a hot tip, softened plastic, energized part, or unventilated fumes.

10. Pre-Use Checklist

Before using the heat insert tool, the operator shall:

1. Confirm they are trained and not impaired, distracted, or rushed.
2. Confirm the bench is clean, stable, dry, and free of unnecessary combustible material.
3. Inspect the tool, controller, stand, holder, tip, fixture, power cord, plug, and temperature controls for damage.
4. Confirm fume extraction or local ventilation is operating when required.
5. Confirm the part material, insert, hole size, tip, and fixture are approved for the process.
6. Confirm the selected temperature and dwell time are appropriate.
7. Confirm the workpiece is not powered and does not contain a battery, sealed component, or pressure-sensitive item.
8. Confirm the fire extinguisher is accessible.
9. Confirm safety glasses and other required PPE are worn.
10. Confirm tools are available for safe holding, insertion, and hot-part handling.

If anything is uncertain, stop and do not operate the equipment until the safe setup and procedure are clear.

11. Operating Rules

Operators shall follow these rules during every heat insert task:

- Use only insert tips, inserts, fixtures, and materials approved for the tool and part.
- Set the lowest practical temperature that reliably installs the insert without charring, smoking, cracking, or deforming the part.
- Keep the hot tool in its stand or press holder whenever it is not actively in use.
- Never place a hot tool directly on the bench.
- Keep hands, sleeves, cords, and tools clear of the hot tip and press path.
- Support the part securely. Do not hold a small part in a way that places fingers under the insert tip or press head.
- Apply steady controlled force. Do not force an insert if it binds, tilts, cracks the part, or requires unsafe pressure.
- Do not touch inserts, tips, surrounding plastic, or fixtures until they have cooled or proper heat protection is used.
- Use fume extraction when heating plastics or any material that produces odor or fumes.
- Do not leave a powered heat insert tool unattended.
- Do not use damaged cords, cracked handles, loose tips, unstable stands, or malfunctioning controllers.
- Turn down, sleep, or power off the tool during long pauses according to the lab procedure.

Stop immediately if plastic chars, fumes are not captured, the tool overheats, the handle or cord smokes, the part becomes unstable, or the process behaves differently than expected.

12. Shutdown and Post-Use Procedure

After using the heat insert tool, the operator shall:

1. Place the tool securely in its stand, holder, or press.
2. Power off the tool unless another trained operator is immediately taking over.
3. Allow the tip, stand, fixtures, inserts, and processed parts to cool before handling or storage.
4. Continue ventilation as needed until fumes and odors are cleared.
5. Return inserts, tips, fixtures, tools, chemicals, and materials to their proper storage locations.
6. Dispose of failed parts, contaminated wipes, sharp debris, and chemical waste properly.
7. Clean the bench without spreading dust, residue, or debris into the room.
8. Wash hands after cleanup.

Do not store a hot tool, hot fixture, hot insert, or hot plastic part in a drawer, bin, bag, cabinet, or near combustible material.

13. Maintenance and Troubleshooting

Routine maintenance may be performed only by trained users. Non-routine maintenance, electrical repair, controller repair, calibration, press adjustment beyond normal setup, heating-element replacement, or cord replacement must be performed only by a person competent for that task under a specific safe procedure.

Before maintenance, the operator shall:

- Power off and unplug equipment when required.
- Allow hot components to cool unless heat is required by the approved procedure.
- Use proper tools and PPE.
- Keep liquids away from electrical parts.

During troubleshooting:

- Never touch exposed energized parts.
- Never bypass grounding, fuses, thermostats, timers, or temperature controls.
- Never operate the tool with a damaged cord, cracked handle, loose tip, unstable stand, unreliable temperature control, or suspected electrical fault.
- Tag unsafe equipment out of service and report it.

14. Housekeeping and Waste

The heat insert area must be kept clean and orderly.

Operators shall:

- Keep the area free of loose paper, cardboard, packaging, wipes, foam, plastic bags, and unnecessary combustibles.
- Keep sharp inserts, failed parts, broken plastic, and sharp fixtures contained.
- Keep cleaners, solvents, adhesives, and chemicals capped and labeled.
- Keep wet items and liquids away from powered equipment.
- Dispose of contaminated wipes, adhesive waste, chemical waste, sharp debris, and failed parts according to lab procedures.

Do not blow dust, dried residue, plastic debris, or part debris into the room with compressed air.

15. Emergency Procedures

15.1 Burns

- Move away from the hot source.
- Cool the burn with clean cool water according to first-aid procedure.
- Do not pull hardened plastic or other stuck material from skin.
- Report the injury immediately.

- Seek medical attention for serious burns, burns involving molten or softened material, electrical burns, or burns larger than a minor superficial injury.

15.2 Eye Injury

- Stop work immediately.
- Do not rub the eye.
- Use the eyewash station if chemical exposure or particles are suspected and it is appropriate to do so.
- Seek medical attention for insert impact, plastic fragments, embedded particles, chemical splash, or persistent irritation.
- Report the incident immediately.

15.3 Cuts, Punctures, or Pinch Injury

- Stop work.
- Use first aid as appropriate.
- Seek medical attention for deep cuts, punctures from contaminated items, embedded debris, crush injury, uncontrolled bleeding, or loss of motion or sensation.
- Report the injury immediately.

15.4 Fume Exposure or Ventilation Failure

- Stop heating immediately if safe to do so.
- Move away from the fumes.
- Turn off the tool if safe to do so.
- Notify the person responsible for the lab or equipment.
- Do not resume until ventilation is restored and the cause is understood.
- Seek medical attention for breathing difficulty, chest tightness, severe irritation, dizziness, headache, nausea, or persistent symptoms.

15.5 Fire, Smoke, or Burning Smell

- Stop work immediately.
- Power off or unplug the tool only if safe to do so.
- Alert nearby personnel.
- Use a fire extinguisher only if trained, the fire is small, and there is a clear exit path.
- Evacuate and call emergency services if the fire grows, produces heavy smoke, involves electrical equipment, involves chemicals, or cannot be controlled immediately.
- Do not resume operation until the equipment and work area have been inspected and the cause has been corrected.

15.6 Electrical Fault

- Stop work.
- Do not touch exposed wiring, damaged cords, wet equipment, or energized parts.
- Disconnect power only if safe to do so.
- Tag the equipment out of service.
- Notify the person responsible for the lab or equipment.

16. Stop-Work Authority

Every employee has authority to stop heat insert work if they believe conditions are unsafe. Work may resume only when the unsafe condition is corrected and the person handling the equipment has confirmed the setup is safe.

17. References for Safety Basis

- Soldering Iron Safety Policy in this repository, especially the controls for hot tools, fumes, fire prevention, electrical safety, and tool stands.
- Bambu Lab P2S 3D Printer Safety Policy in this repository, especially the controls for heated plastics, 3D-printed parts, fumes, sharp plastic, part removal, and housekeeping.
- Manufacturer manuals, labels, specifications, temperature limits, maintenance instructions, and process guidance for the exact heat insert tool, heat-set insert press, tips, fixtures, and fume extraction equipment used in the lab.
- Safety data sheets and supplier information for heat-set inserts, plastics, adhesives, cleaners, solvents, and other materials used in the lab.
- OSHA general industry safety principles for personal protective equipment, hazard communication, electrical safety, fire prevention, machine guarding, and safe work practices.
- NIOSH and other occupational health guidance regarding local exhaust ventilation and heated plastic emissions.

18. Acknowledgement

By signing below, I acknowledge that:

- I have read and understood this Heat Insert Tool Safety Policy.
- I understand the ways heat insert tools can injure me or others, including burns, softened material contact, eye injury, fumes, fire, electrical shock, cuts, punctures, pinch injury, and ergonomic strain.
- I agree to operate heat insert tools only when trained, unimpaired, and following the safe procedure.
- I agree not to leave hot tools unattended, bypass safety controls, heat unsuitable materials, use damaged equipment, skip required ventilation, or allow untrained operation.

- I agree to report incidents, near misses, equipment faults, unsafe conditions, smoke, fire, abnormal odors, and injuries immediately.

Signature

Date